Exercise 6
Differential Amplifiers

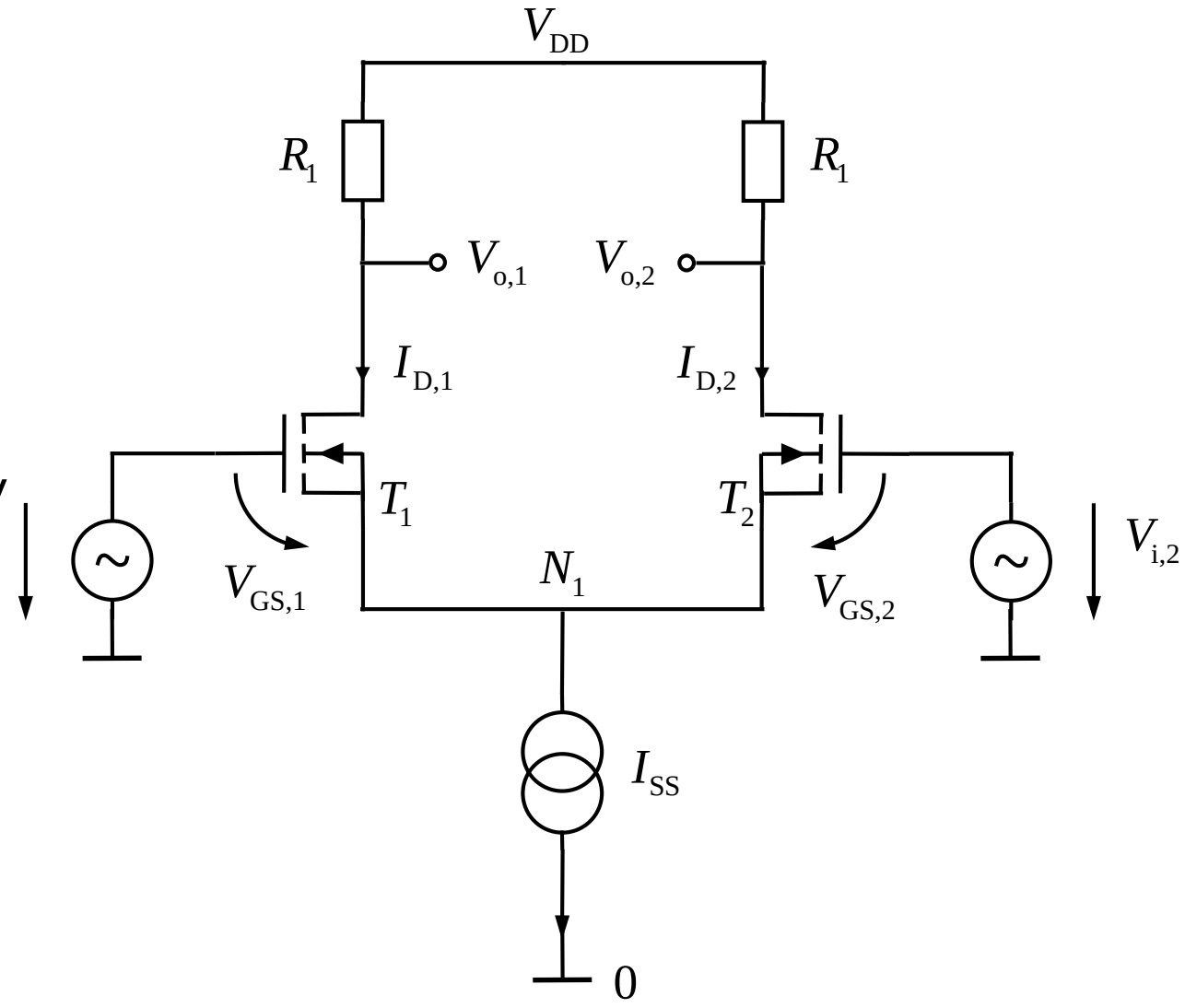
Winter Term 2025/26

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Exercise 6 – DA

Specifications:

- T1=T2
- V_{DD}= 1.2V
- I_{SS} = 100 µA
- Input/output common mode = 0.8 V
- β_0 = 360 µA/V²
- λ = 0.022
- V_{th}= 188 mV
- γ = 0.1 $V^{0.5}$
- V_{SB} = 0 V



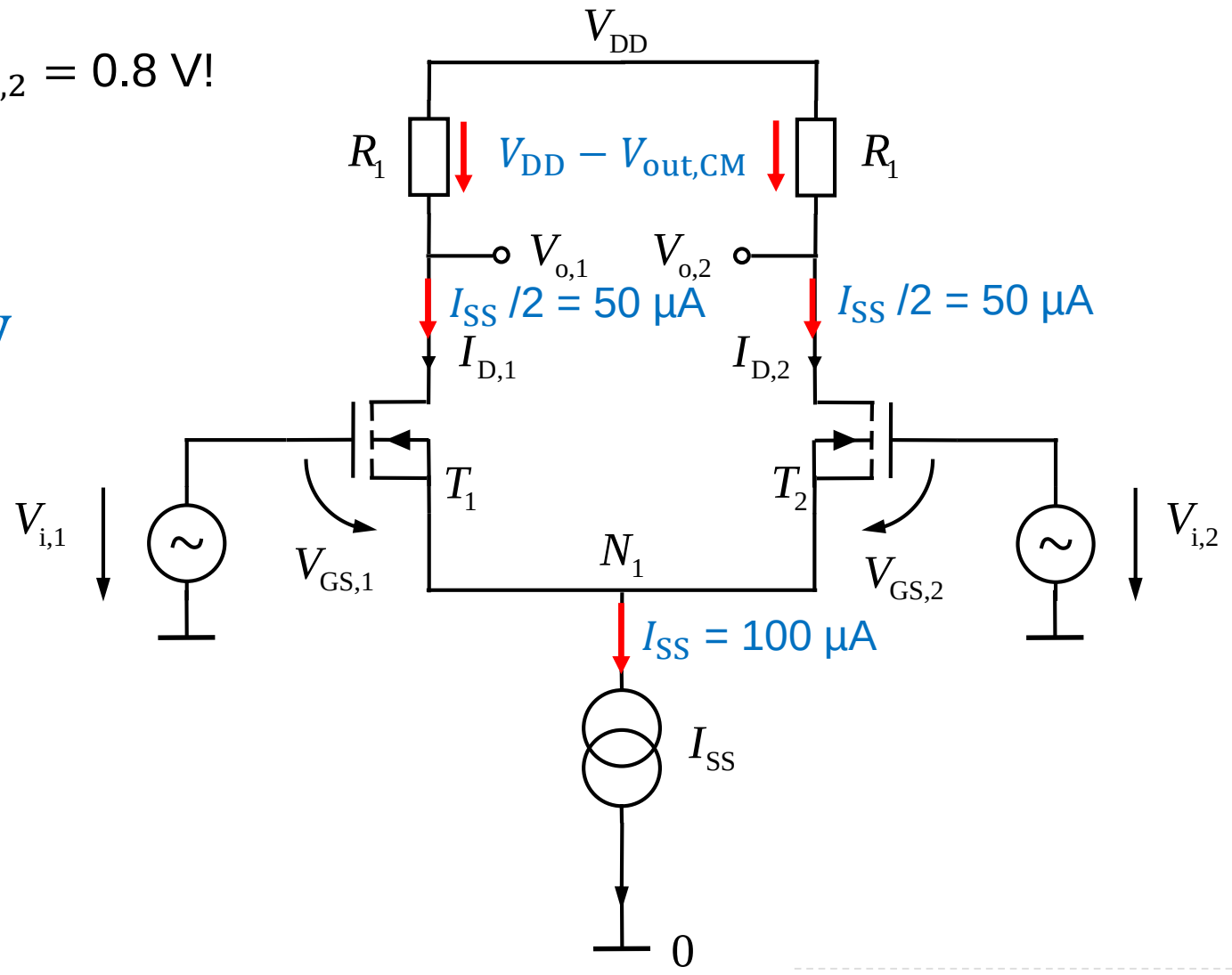
Exercise 6.1 – DA

1. Find the dimension of R_1 for $V_{o,1} = V_{o,2} = 0.8$ V!

The circuit is symmetrical!

- $I_{R1} = \frac{I_{SS}}{2} = 50\ \mu A$
- $V_{R1} = V_{DD} - V_{out,CM} = 0.4\ V$
- $R_1 = \frac{V_{R1}}{I_{R1}} = \frac{0.4\ V}{50\ \mu A}$

- $\boldsymbol{R_1 = 8\ kOhm}$



Exercise 6.2 – DA

2. Find the width of the input transistors for L = 10 µm and V_{GSeff} = 162 mV! The channel-length modulation is to be neglected!

Remember: All transistors must be kept in saturation!

$$\frac{I_{\mathrm{SS}}}{2} = \frac{\beta_0}{2}\frac{W_{1,2}}{L_{1,2}}\left(V_{\mathrm{GS1,2}} - V_{\mathrm{th}}\right)^2$$

$$\rightarrow\ W_{1,2} = \frac{2 \cdot L_{1,2} \cdot I_{\mathrm{D1,2}}}{2 \cdot \beta_0 \cdot \left(V_{\mathrm{GS1,2}} - V_{\mathrm{th}}\right)^2}$$

$$W_{1,2} = \frac{2 \cdot 10\ \mu\mathrm{m} \cdot 100\ \mu\mathrm{A}}{2 \cdot 360\ \frac{\mu\mathrm{A}}{\mathrm{V}^2} \cdot (162\ \mathrm{mV})^2}$$

$$W_{1,2} \approx \mathbf{106\ \mu m}$$

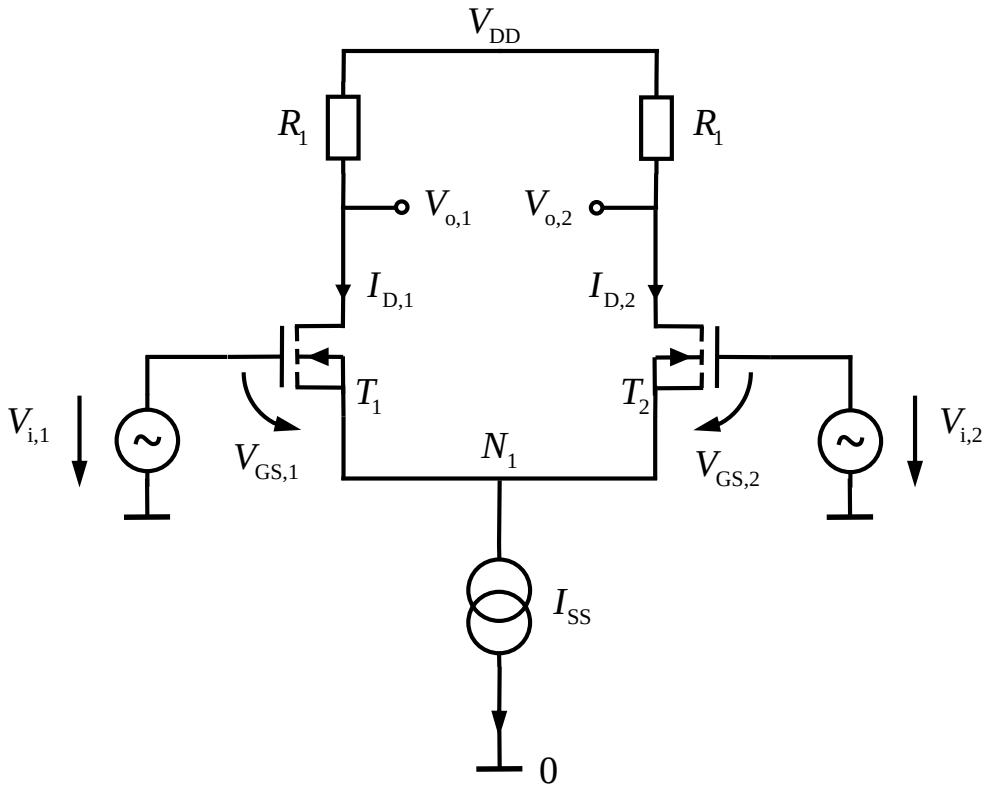


Figure 1: Differential amplifier

Exercise 6.3 – DA

3. Determine the **input common-mode range**! The channel-length modulation is to be neglected!

- Calculation of the $\boldsymbol{V_{\mathrm{CM,in,min}}}$:
- Transistors T_1 and T_2 must be turned on & biased in saturation

Assuming that $V_{\mathrm{N1}} \geq 0$ at all time

- $V_{\mathrm{GS1,2}} = V_{\mathrm{GSeff1,2}} + V_{\mathrm{th}} = 0.35\ \mathrm{V}$
- $V_{\mathrm{CM,in,min}} = V_{\mathrm{N1}} + V_{\mathrm{GS1,2}}$

- The lowest possible input potential

is given for $V_{\mathrm{N1}} = 0\ \mathrm{V}$

- $V_{\mathrm{CM,in,min}} = 0\ \mathrm{V} + 0.35\ \mathrm{V}$
- $\boldsymbol{V_{\mathrm{CM,in,min}} = 0.35\ \mathrm{V}}$

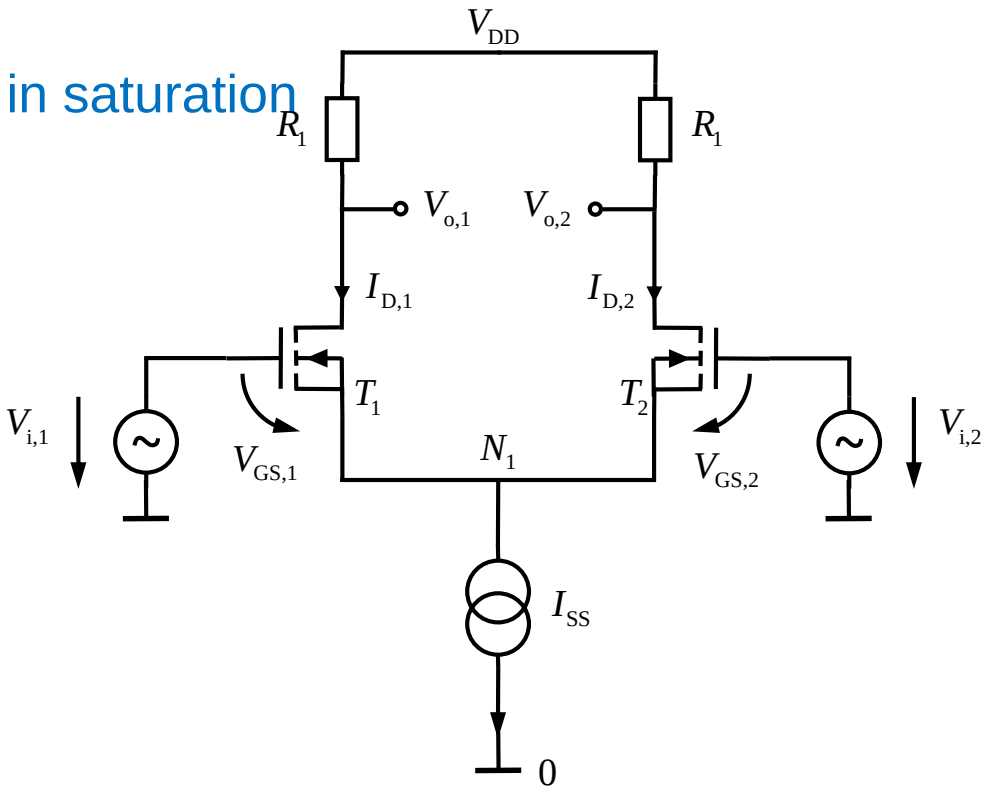


Figure 1: Differential amplifier

Exercise 6.3 – DA

3. Determine **the input common-mode range**! The channel-length modulation is to be neglected!

- Calculation of the $\mathbf{V_{CM,in,max}}$:
- Transistors T_1 and T_2 must be turned on & biased in saturation
  - $V_{DS1,2} \geq V_{GS1,2} - V_{th} \leftrightarrow V_{D1,2} \geq V_{CM,in,max} - V_{th}$
- $V_{D1,2}$ is determined by design
  - $V_{D1,2} = V_{o1,2} = V_{DD} - R_1 \cdot \frac{I_{SS}}{2} = 0.8\,V$
- Therefore
  - $V_{CM,in,max} = V_{D1,2} + V_{th}$
  - $V_{CM,in,max} = 0.8\,V + 0.188\,V$
  - $\mathbf{V_{CM,in,max} = 0.988\,V}$

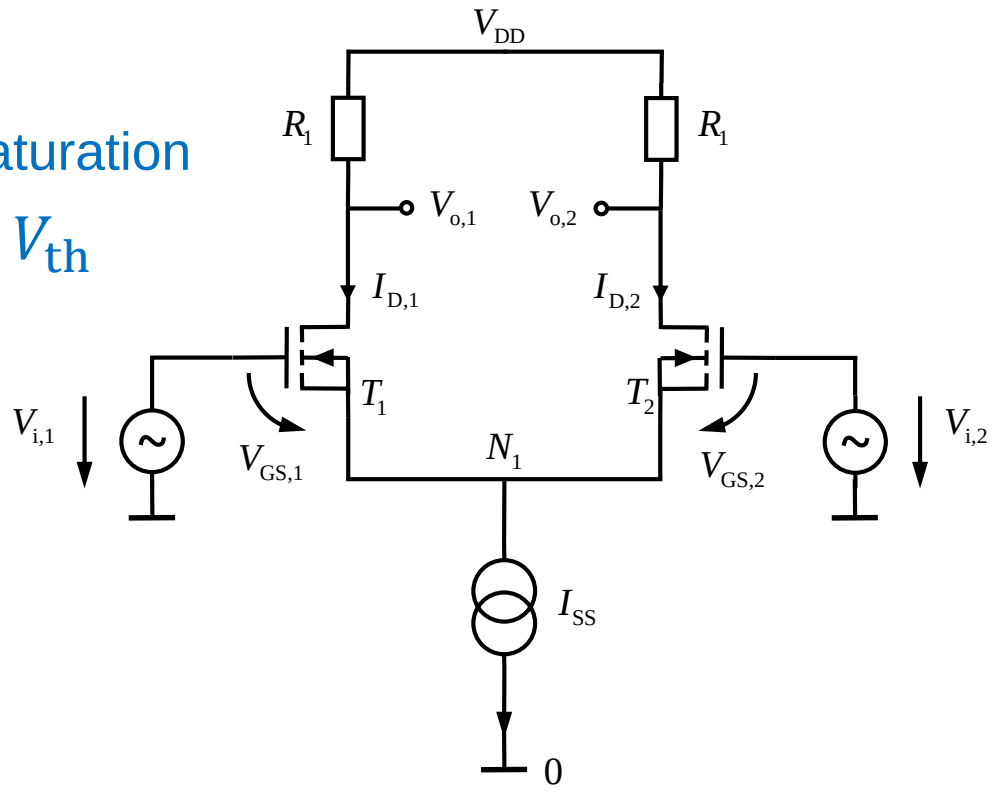


Figure 1: Differential amplifier

Exercise 6.3 – DA

3. Determine the input common-mode range! The channel-length modulation is to be neglected!

- $V_{\mathrm{CM,in}} = V_{\mathrm{CM,in,max}} - V_{\mathrm{CM,in,min}}$
- $V_{\mathrm{CM,in}} = 0.988\ \mathrm{V} - 0.35\ \mathrm{V}$
- $\boldsymbol{V_{\mathrm{CM,in}} = 0.638\ \mathrm{V}}$

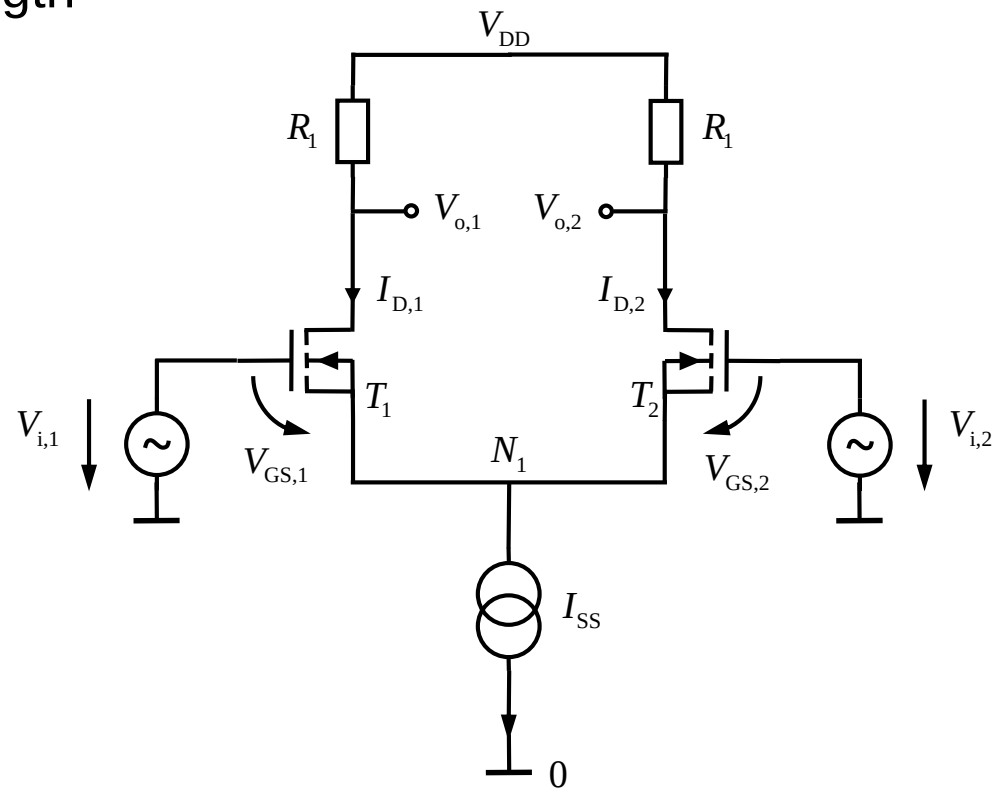


Figure 1: Differential amplifier

Exercise 6.4 – DA

4. Replace the ideal current source with an NMOS transistor labeled T_{SS}! Dimension the width of the transistor T_{SS} for L = 10 µm and V_{GSeff} = 162 mV!

- T_{SS} exhibits the same length $\mathrm{L} = 10\ \mu\mathrm{A}$ and the same effective gate source voltage $V_{GS,eff} = 162\ \mathrm{mV}$ as the input transistors T_1 and T_2.
- The current I_{SS} flowing through T_{SS} is twice as large as the currents flowing through T_1 and T_2.
- The width of T_{SS} thus results in
  - $W_{SS} = 2 \cdot W_{1,2}$
  - $W_{SS} = 2 \cdot 106\ \mu\mathrm{m}$
  - $\boldsymbol{W_{SS} = 212\ \mu\mathrm{m}}$

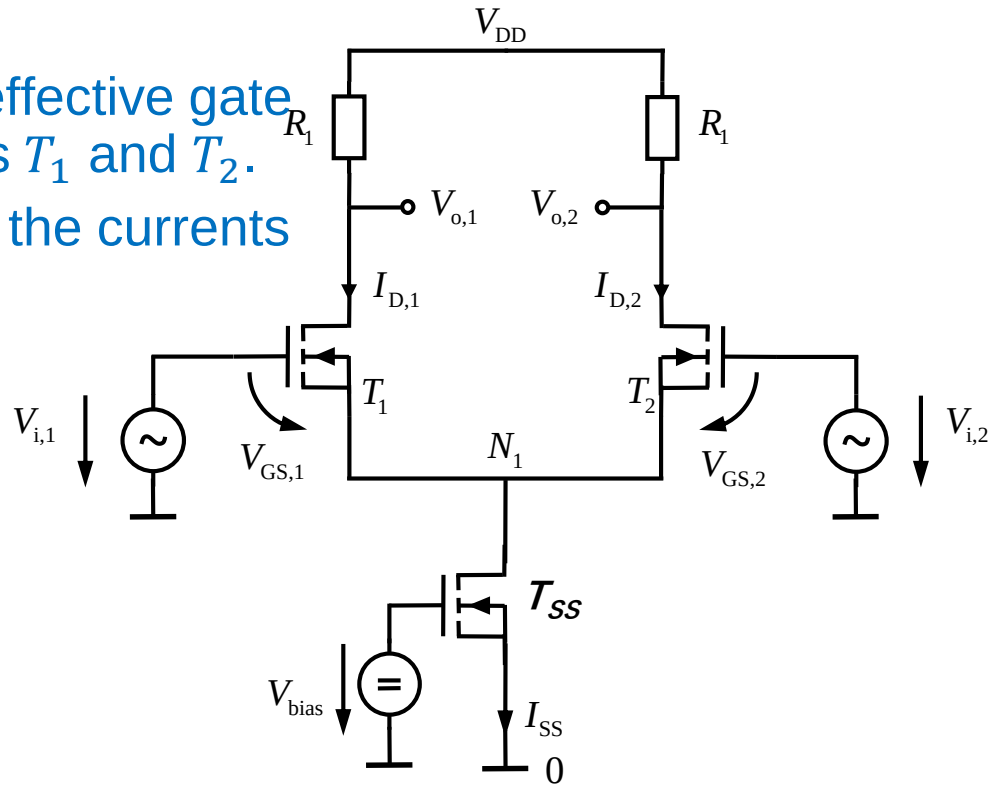


Figure 2: Differential amplifier

Exercise 6.5 – DA

5. How large is the input common mode range now? The channel-length modulation is to be neglected.

- $V_{\mathrm{CM,in,max}}$ → unchanged

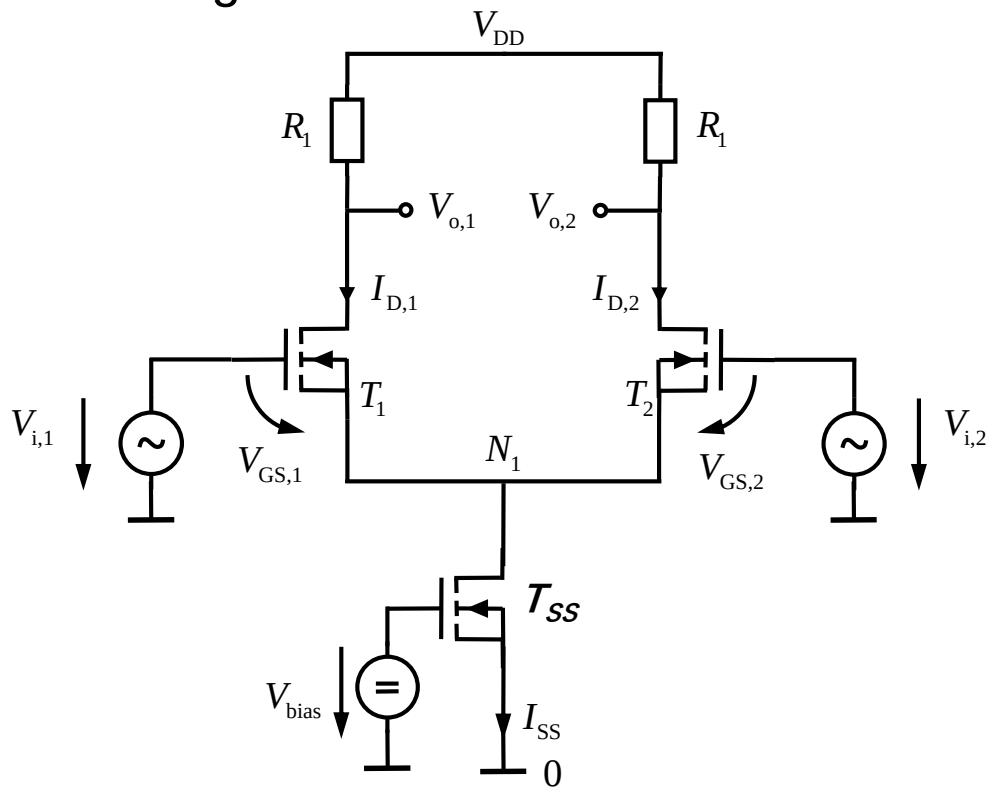


Figure 2: Differential amplifier

Exercise 6.5 – DA

5. How large is the input common mode range now? The channel-length modulation is to be neglected.

- The source potentials of the transistors $T_{1,2}$ are the drain potential of the current source transistor T_{SS}
- The current source T_{SS} must be operated in **saturation**
  - $V_{DS,SS} = V_{N1} = V_{CM,in,min} - V_{GS1,2}$
  - $V_{CM,in,min} - V_{GS1,2} \geq V_{GS,eff}$
  - $V_{CM,in,min} \geq V_{GS1,2} + V_{GS,eff}$
  - $V_{CM,in,min} \geq 0.35\ \text{V} + 0.162\ \text{V}$
  - $\boldsymbol{V_{CM,in,min} \geq 0.512\ \text{V}}$
- $\boldsymbol{V_{CM,in,min}}$ increases by $\boldsymbol{V_{GS,eff} = 0.162\ \text{mV}}$

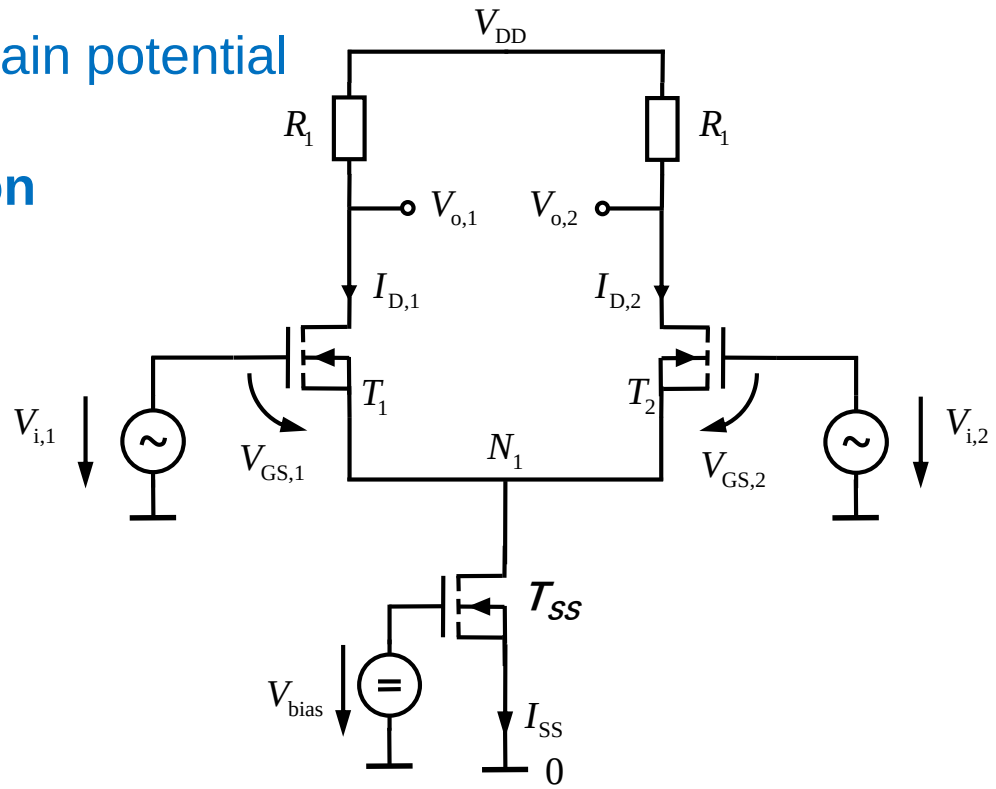


Figure 2: Differential amplifier

Exercise 6.5 – DA

6. How large is the input common mode range now? The channel-length modulation is to be neglected.

- $V_{\mathrm{CM,in,nmosCS}} = V_{\mathrm{CM,in,max}} - (V_{\mathrm{CM,in,min,idealCS}} + V_{\mathrm{GS,eff}})$
- $V_{\mathrm{CM,in,nmosCS}} = 0.988\ \mathrm{V} - (0.512\ \mathrm{V} + 0.162\ \mathrm{V})$
- $\boldsymbol{V_{\mathrm{CM,in,nmosCS}} = 0.476\ \mathrm{V}}$
- $V_{\mathrm{CM,in}}$ is reduced by $V_{\mathrm{GS,eff}} = 0.162\ \mathrm{mV}$
  - $V_{\mathrm{CM,in,nmosCS}} = V_{\mathrm{CM,in,idealCS}} - 0.162\ \mathrm{V}$

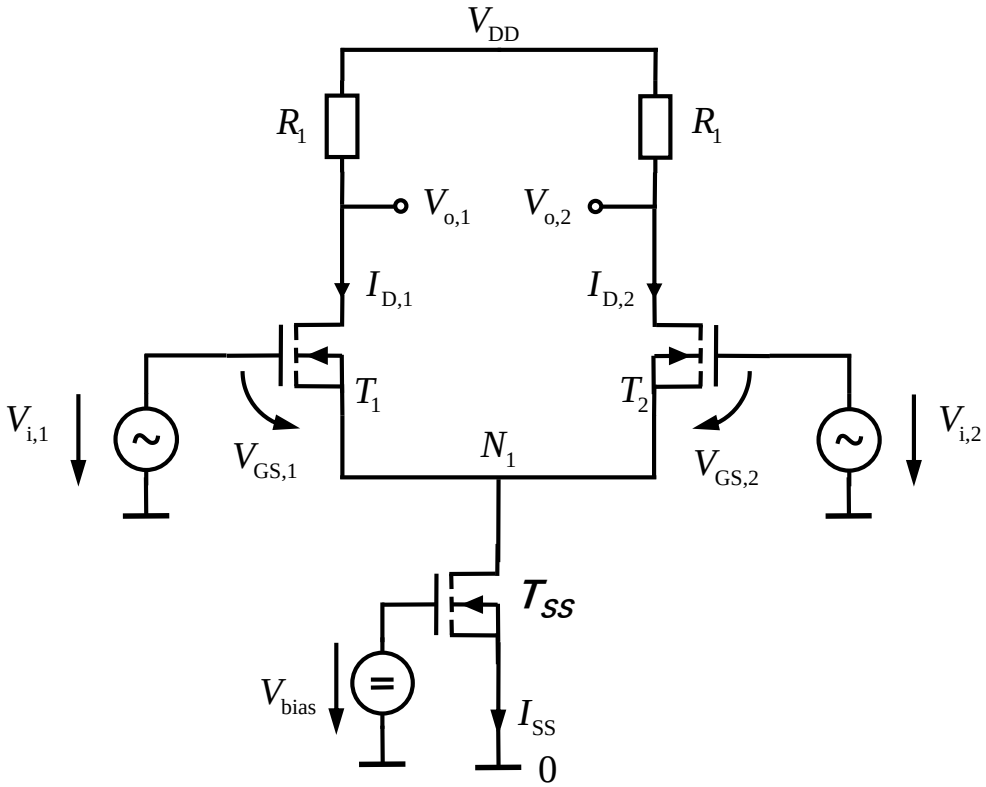


Figure 2: Differential amplifier

Exercise 6.6 – DA

6. Briefly explain how the input common mode range changes when the bulks of the input transistors are connected to ground, i.e., $V_{SB} \neq 0$ V!

- Due to the body effect, the threshold voltage is increased:
  - $V_{SB} \neq 0 \leftrightarrow V_{th}(V_{SB}) \approx V_{th0} + \gamma \cdot \sqrt{V_{SB}}$
- $V_{in1,2}$ low → $V_{SB,1,2}$ low → ΔV_{th} low
- $V_{in1,2}$ high → $V_{SB,1,2}$ high → ΔV_{th} high

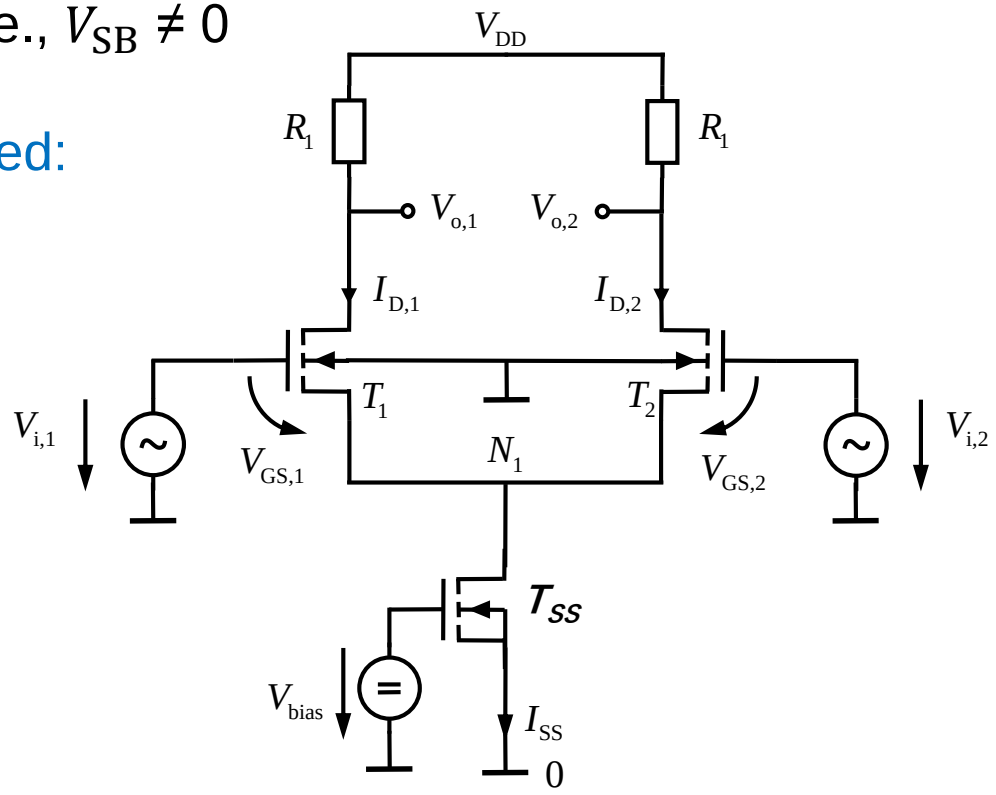


Figure 3: Differential amplifier

Exercise 6.6 – DA

6. Briefly explain how the input common mode range changes when the bulks of the input transistors are connected to ground, i.e., $V_{SB} \neq 0$ V!

- $V_{in1,2} = V_{CM,in,min}$
  - $V_{SB1,2} = V_{N1}$
- From task 3 → $V_{N1}(V_{CM,in,min}) = V_{GS,eff}$
  - $V_{th}(V_{SB} = 0.162\,\text{V}) = 0.188 + 0.1 \cdot \sqrt{0.162}$
  - $V_{th}(V_{SB} = 0.162\,\text{V}) = 0.228\,\text{V}$
- $V_{CM,in,min}$ is calculated similarly to task 3
  - $V_{CM,in,min} = V_{GS1,2} + V_{GS,eff}$
  - $V_{CM,in,min} = V_{th}(V_{SB} = 0.162\,\text{V}) + 2 \cdot V_{GS,eff}$
  - $V_{CM,in,min} = 0.228\,\text{V} + 2 \cdot 0.162\,\text{V}$
  - $\mathbf{V_{CM,in,min} = 0.552\,V}$

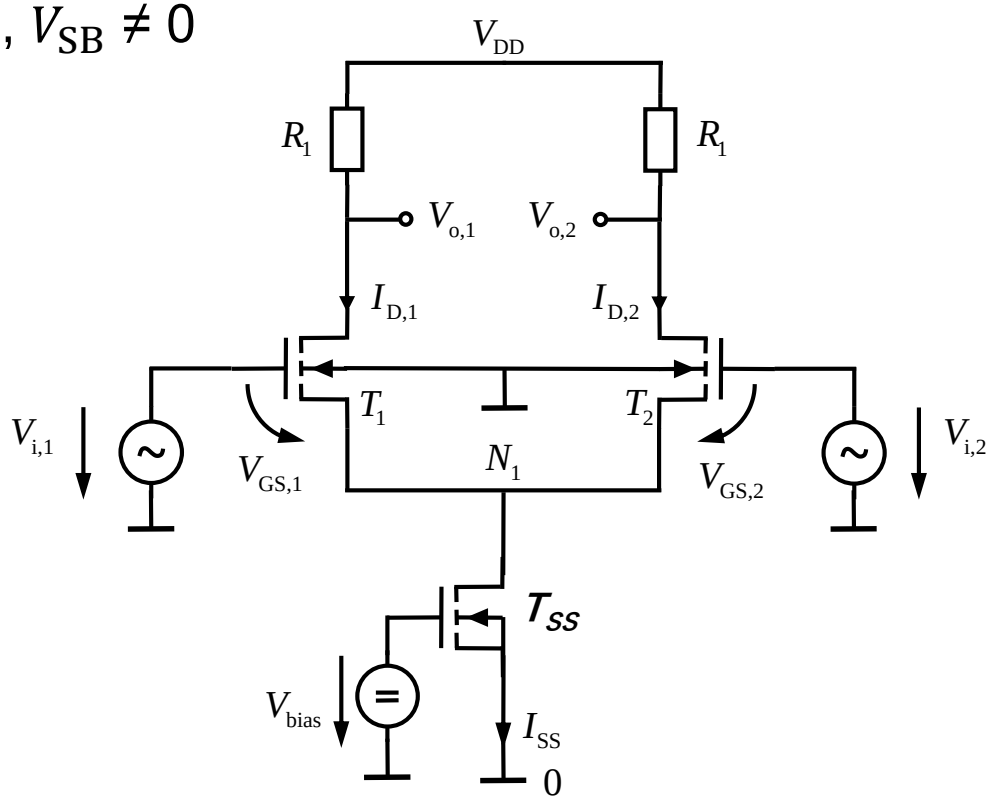


Figure 3: Differential amplifier

Exercise 6.6 – DA

6. Briefly explain how the input common mode range changes when the bulks of the input transistors are connected to ground, i.e., $V_{SB} \neq 0$ V!

- $V_{in1,2} = V_{CM,in,max}$
- $V_{CM,in}$ can be expressed as a function of $V_{S1,2}$
  - $V_{CM,in} = V_{S1,2} + V_{GS1,2}$ (1)
- When $T_{1,2}$ operate at the limit of saturation
  - $V_{DS1,2} = V_{GS1,2} - V_{th}(V_{SB1,2})$
  - $\Rightarrow V_{D1,2} = V_{CM,in} - V_{th}(V_{SB1,2})$ (2)
- Combining (1) and (2) yields
  - $V_{D1,2} = V_{S1,2} + V_{GS1,2} - V_{th}(V_{SB1,2})$
  - $V_{S1,2} = V_{D1,2} - V_{GSeff1,2}$
  - $V_{S1,2} = 0.638\ \text{V}$
- From task 3 we determine $V_{CM,in,max}$
  - $V_{CM,in,max} = V_{D1,2} + V_{th}(V_{SB})$
  - $V_{CM,in,max} = 0.8 + 0.188 + 0.1 \cdot \sqrt{0.638}$
  - $\boldsymbol{V_{CM,in,max} = 1.068\ \text{V}}$

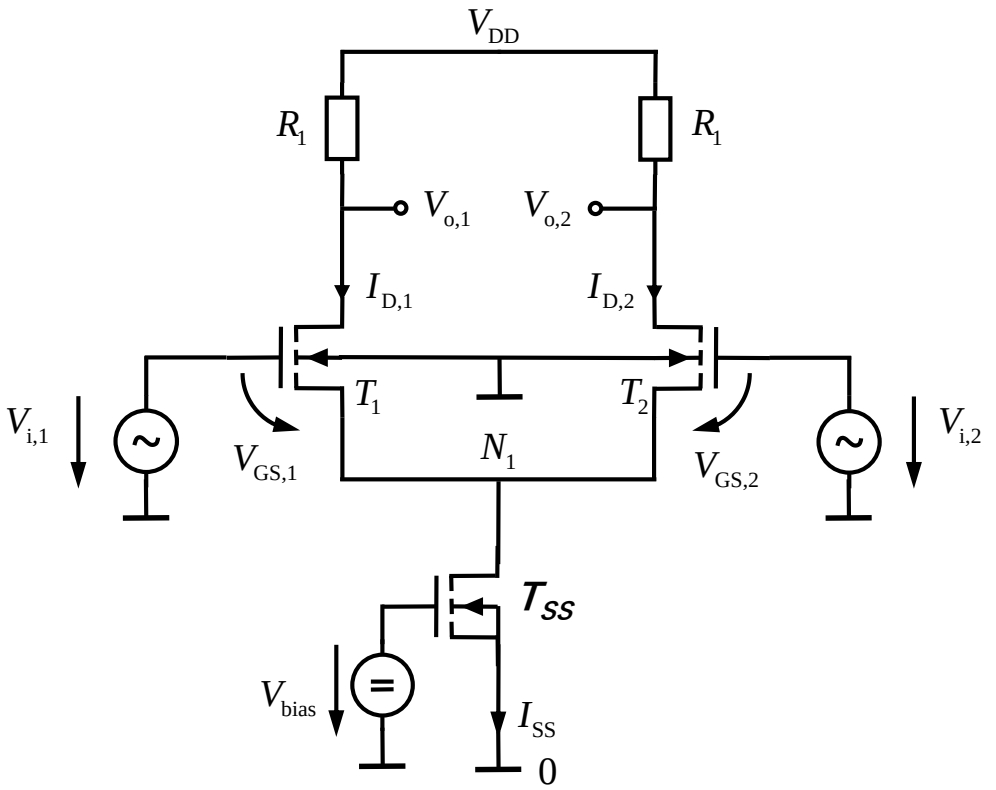


Figure 3: Differential amplifier

Exercise 6.6 – DA

6. Briefly explain how the input common mode range changes when the bulks of the input transistors are connected to ground, i.e., $V_{\mathrm{SB}} \neq 0$ V!

- $V_{\mathrm{CM,in,BE}} = (V_{\mathrm{CM,in,max},old} + 80\ mV) - (V_{\mathrm{CM,in,min,old}} + 40\ mV)$
- $V_{\mathrm{CM,in,BE}} = V_{\mathrm{CM,in},old} + 40\ mV$
- $\boldsymbol{V_{\mathrm{CM,in,BE}} = 0.516\ \mathrm{V}}$

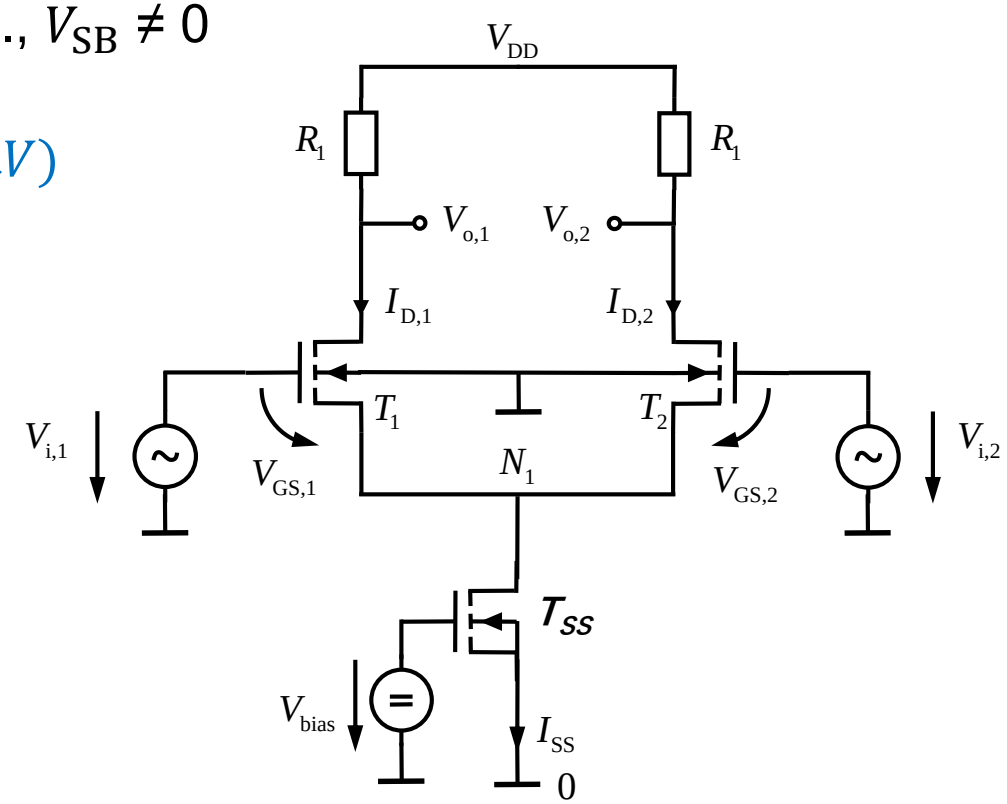


Figure 3: Differential amplifier

Exercise 6.7 – DA

7. Implement and simulate the differential stage in LTspice for $V_{SB} = 0\ V$, and the cases “ideal current source” and “transistor-based current source”!

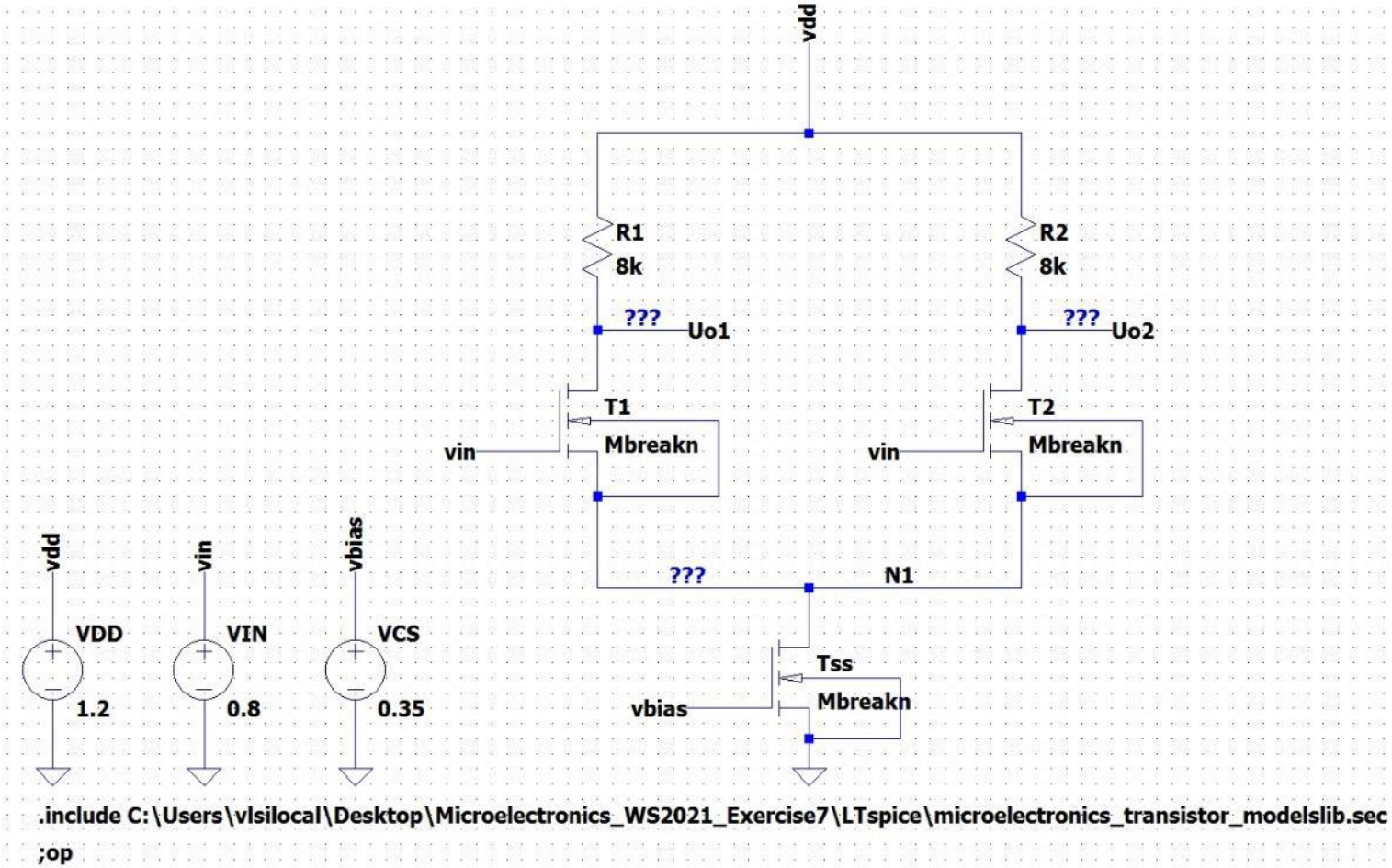


Figure 4: Implementation of the differential stage in LTspice ($V_{SB} = 0\ V$, transistor-based current source).

Exercise 6.7 – DA

7. Implement and simulate the differential stage in LTspice for $V_{\mathrm{SB}} = 0\ \mathrm{V}$, and the cases “ideal current source” and “transistor-based current source”!

- Operating point simulation
  - .op

	specifications / calculations	ideal current source	NMOS current source
$V_{\mathrm{o1,2}}$ (mV)	800.0	800.0	786.8
V_{N1} (mV)	400.0	403.7	401.2
I_{SS} (µA)	100.0	100.0	103.3
$I_{\mathrm{D1,2}}$ (µA)	50.0	50.0	51.6
$V_{\mathrm{GS1,2}}$ (mV)	350.0	396.3	398.8

Exercise 6.7 – DA

7. Implement and simulate the differential stage in LTspice for $V_{SB} = 0\ V$, and the cases “ideal current source” and “transistor-based current source”!

- ICMR simulation
  - .dc VIN 0 1.2 1m

	ideal current source		NMOS current source	
	calculation	simulation	calculation	simulation
$V_{CM,in,min}$ (mV)	350	347	512	507
$V_{CM,in,max}$ (mV)	988	988	988	972
$V_{CM,in}$ (mV)	638	641	476	465